

Teaching Guide: 21 Oncology AI Tools

Access Model, Cost, and How to Teach the Clinical Practice/Day-to-Day Relevance of Each

Important correction before we start: not all 21 tools are free. Roughly a third are genuinely free or open-source; the rest are commercial, hospital-licensed enterprise software that individual students cannot access without an institutional deployment. This document is honest about which is which — teaching students that a \$2M hospital-licensed radiotherapy planning system is “free to try” would set them up for a bad surprise on their first day of residency.

How this maps to your Virtual Clinical Sandbox (VCS) philosophy: for the genuinely free/open tools, students can get real hands-on practice. For the enterprise-only tools, the correct teaching method is the same one your core course already uses for hospital-only tools — case studies, published validation data, and screenshots/demo videos, not hands-on access.

🌀 GENUINELY FREE / OPEN — Hands-On Lab Activities Possible

1. AlphaFold3 (via AlphaFold Server)

- **Access:** Free for academic/non-commercial use at alphafoldserver.com, with some restrictions on certain ligand types for commercial use
- **Clinical practice relevance:** Structural biology underlies “why does this mutation matter” — when a student sees a variant report, AlphaFold’s predicted structure is what an oncologist can point to when explaining why a specific mutation changes drug binding
- **How to teach it:** Live demo — enter a real (public, non-patient) protein sequence, watch it fold in real time. Emphasize this is upstream research infrastructure, not something used at the bedside directly
- **Day-to-day importance:** Low direct daily use for a clinician, high importance for understanding why a targeted therapy was chosen for a specific mutation

2. DeepVariant (Google/Broad Institute)

- **Access:** Fully free, open-source, on GitHub
- **Clinical practice relevance:** This is very likely the invisible engine behind the tumor NGS report a student will read — most modern genomic sequencing pipelines use a variant caller like this one under the hood
- **How to teach it:** Don’t have students run it (it’s a bioinformatics pipeline tool, not a clinical UI) — instead, teach it as “the thing that makes your mutation report possible,” then show a real de-identified sample NGS report and trace which fields came from variant-calling
- **Day-to-day importance:** Zero direct interaction, but foundational — every time a student reads “EGFR L858R detected,” a tool like this made that call

3. MatchMiner (Dana-Farber)

- **Access:** Fully free, open-source, publicly available
- **Clinical practice relevance:** Matches a patient’s genomic profile to open clinical trials automatically — directly relevant the moment a student’s patient has exhausted standard therapy and trial enrollment becomes the next option
- **How to teach it:** Hands-on lab activity — load a synthetic patient’s genomic profile (from Synthea or a public dataset) and have students run a mock trial-matching search
- **Day-to-day importance:** High for oncology-track students — this is a task residents are frequently asked to do manually today

4. TrialMatchAI

- **Access:** Open-source, code publicly available (research tool, requires some technical setup to deploy)
- **Clinical practice relevance:** Same use case as MatchMiner — trial matching — but demonstrates the newer LLM-based approach (processes unstructured physician notes directly, not just structured fields)
- **How to teach it:** Best as an instructor-led demo rather than student hands-on, given the setup complexity — show the before/after of feeding it a messy clinical note versus a structured field
- **Day-to-day importance:** Same high relevance as MatchMiner, but illustrates the future direction (note-based matching) rather than today’s standard

5. ChatGPT for Clinicians (OpenAI)

- **Access:** Free for verified physicians, NPs, PAs, or pharmacists (launched April 2026)

- **Clinical practice relevance:** The most likely tool a student will actually open on their own phone between patients — but the JMIR study found only “moderate but meaningful alignment” with UpToDate, strongest in pharmacology
- **How to teach it:** Hands-on — have students ask it the same clinical question they’d ask UpToDate, then compare answers side by side. This is the best possible lab exercise for teaching “accessible adjunctive tool, not a substitute”
- **Day-to-day importance:** Very high adoption likelihood, which is exactly why the verification habit matters most here

6. OpenEvidence

- **Access:** Free for verified healthcare practitioners
- **Clinical practice relevance:** Official partner of NEJM, JAMA, and Cochrane — this is the “replace your unread bookmark folder” tool from the Sermo article
- **How to teach it:** Hands-on — have students generate a 2-3 minute evidence digest on a topic in their specialty and present it to the class
- **Day-to-day importance:** High — directly targets the documented physician pain point of literature staying unread

🌀 FREE FOR RESEARCH / ACADEMIC USE ONLY — Teach as Case Study

7. Chai Discovery (Chai-1, older open-weight version)

- **Access:** The earlier Chai-1 model is open-weight for academic use; the current commercial Chai-3 is not
- **Clinical practice relevance:** Illustrates the drug-discovery pipeline a student’s future patients’ medications came from — not something a clinician uses directly
- **How to teach it:** Case study only — walk through one real molecule’s journey from AI-designed candidate to Phase I trial (the deck already has this: Rentosertib, now in Phase III)
- **Day-to-day importance:** Zero direct clinical use; high value for understanding where new drugs come from

8. COMPASS (Harvard Medical School)

- **Access:** Published research model (Nature Medicine 2026) — not yet a deployed clinical product; code/data typically available for academic replication
- **Clinical practice relevance:** Predicts immunotherapy response from tumor gene expression — this is the future of “should I give this patient a checkpoint inhibitor?”
- **How to teach it:** Case study — walk through the actual published performance numbers (8.5% improvement over prior methods) and discuss what would need to happen before this reaches a bedside product
- **Day-to-day importance:** Not usable today, but a strong discussion prompt for “what’s coming next”

9. SCORPIO (Mount Sinai / MSK)

- **Access:** Research tool, not a commercial product yet
- **Clinical practice relevance:** Predicts immunotherapy response from routine blood tests instead of expensive genomic sequencing — a genuine equity/access story
- **How to teach it:** Case study, paired with a discussion question: “why does using blood tests instead of genomic sequencing matter for patients in lower-resource settings?”
- **Day-to-day importance:** Not usable today; excellent for teaching the health-equity angle of AI in oncology

10. I3LUNG multimodal model

- **Access:** International academic research trial (2,396 patients), not a deployed product
- **Clinical practice relevance:** Demonstrates that AI SUPPORT improved physician accuracy from 0.72 to 0.87 — a rare, well-documented example of human+AI outperforming either alone
- **How to teach it:** This is your best “HITL success story” case study in the whole set — use it explicitly to show that the goal isn’t AI replacing judgment, it’s AI improving a human’s judgment
- **Day-to-day importance:** Not usable today; the single best teaching example for the entire course’s core thesis

11. PRISM-GC (Hebei Medical University)

- **Access:** Active clinical trial (NCT07401199), not a public product
 - **Clinical practice relevance:** Predicting neoadjuvant chemo-immunotherapy response before treatment starts
 - **How to teach it:** Brief mention as an example of an ongoing trial students could watch for results from — good for teaching “how to stay current between board cycles” (ties back to Curso 6 content)
 - **Day-to-day importance:** Not usable today
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● ENTERPRISE / INSTITUTIONAL LICENSE ONLY — Teach via Case Study & Published Data

12. PandaOmics (Insilico Medicine)

- **Access:** Enterprise pharma R&D platform — not available to individual clinicians or students
- **Clinical practice relevance:** None directly — this is upstream research infrastructure
- **How to teach it:** One-slide mention only; the point is awareness, not usage
- **Day-to-day importance:** None for a practicing clinician

13. Triomics

- **Access:** Enterprise deployment (currently at MSK and a few other centers) — requires institutional contract
- **Clinical practice relevance:** Directly relevant to any resident/fellow tasked with trial pre-screening — a 45-minutes-per-patient task this tool compresses dramatically
- **How to teach it:** Case study using MSK's own published deployment data; if your institution ever licenses it, that's when hands-on training happens
- **Day-to-day importance:** High relevance, low personal accessibility — good example of "know this exists so you can advocate for it" rather than "use it yourself"

14. Aidoc

- **Access:** Enterprise hospital software, ~2,000 hospital deployments
- **Clinical practice relevance:** Runs silently in the background flagging urgent findings (e.g., stroke, PE) — most clinicians experience it as an alert, not a tool they open
- **How to teach it:** Case study + a mock alert scenario (e.g., "Aidoc flags a possible intracranial hemorrhage — what do you do first?")
- **Day-to-day importance:** Very high if your hospital has it, zero if it doesn't — good example of "the tool finds you, you don't find it"

15. PathAI (AISight Dx)

- **Access:** Enterprise digital pathology platform
- **Clinical practice relevance:** Directly relevant to pathology-track students; for others, relevant for understanding what's behind a "AI-assisted" note on a pathology report
- **How to teach it:** Case study with published FDA clearance data; pathology-track students should get a screenshot-based walkthrough of the actual interface if available from vendor demo materials
- **Day-to-day importance:** High for pathologists, informational for everyone else

16. ArteraAI Breast

- **Access:** Enterprise diagnostic test, ordered like a lab test, not directly operated by clinicians
- **Clinical practice relevance:** A same-day prognostic score that could change whether a patient gets chemotherapy — directly changes a real treatment conversation
- **How to teach it:** Case study — walk through a realistic scenario: "Your patient's ArteraAI score comes back high-risk. How does this change your conversation with them?"
- **Day-to-day importance:** High relevance for oncology students, ordered rather than operated

17. Ibex Prostate Detect AI

- **Access:** Enterprise digital pathology platform
- **Clinical practice relevance:** Assists Gleason grading — directly relevant to urology/pathology-track students
- **How to teach it:** Case study with the published AUC data (0.905-0.977) — good exercise: "at what AUC would you feel comfortable NOT double-checking a grade yourself?"
- **Day-to-day importance:** High for pathology/urology tracks

18. DoseMeRx

- **Access:** Commercial subscription software
- **Clinical practice relevance:** Directly relevant any time a student is involved in dosing decisions for narrow-therapeutic-index drugs (vancomycin, and increasingly chemotherapy agents)
- **How to teach it:** Case study — walk through one Bayesian dosing calculation conceptually, even without live access to the software
- **Day-to-day importance:** High for pharmacy and inpatient medicine tracks

19. MIM Contour ProtégéAI+ 2.0 (GE HealthCare)

- **Access:** Enterprise radiotherapy planning software
- **Clinical practice relevance:** Directly relevant to radiation oncology-track students — this is literally what most radiation oncologists will touch daily
- **How to teach it:** Case study with the actual FDA clearance summary; if your institution has a radiation oncology rotation partner, request a demo video
- **Day-to-day importance:** Very high for radiation oncology track, low for everyone else

20. AI-Rad Companion Organs RT (Siemens Healthineers)

- **Access:** Enterprise radiotherapy software
- **Clinical practice relevance:** Same use case as #19 — automated organ-at-risk contouring
- **How to teach it:** Case study, paired with #19 as a “compare two vendors” exercise using the published head-to-head validation study
- **Day-to-day importance:** High for radiation oncology track only

21. Varian Smart Segmentation

- **Access:** Enterprise radiotherapy software, built into the Eclipse planning system
- **Clinical practice relevance:** Same use case as #19 and #20 — this is the third vendor in the same competitive space
- **How to teach it:** Complete the three-way vendor comparison started in #19/#20
- **Day-to-day importance:** High for radiation oncology track only

📄 Summary Table: Quick Reference for Lesson Planning

#	Tool	Access	Best Teaching Method	Priority for General (non-oncology-track) Students
1	AlphaFold3	Free (academic)	Live demo	Low
2	DeepVariant	Free (open-source)	Concept only	Low
3	MatchMiner	Free (open-source)	Hands-on lab	High
4	TrialMatchAI	Free (open-source)	Instructor demo	Medium
5	ChatGPT for Clinicians	Free (verified)	Hands-on lab	High
6	OpenEvidence	Free (verified)	Hands-on lab	High
7	Chai Discovery	Free (older model)	Case study	Low
8	COMPASS	Research use	Case study	Medium
9	SCORPIO	Research use	Case study (equity angle)	Medium
10	I3LUNG	Research use	Case study (best HITL example)	High
11	PRISM-GC	Research use	Brief mention	Low
12	PandaOmics	Enterprise	One-slide mention	Low
13	Triomics	Enterprise	Case study	Medium
14	Aidoc	Enterprise	Case study + mock alert	Medium
15	PathAI	Enterprise	Case study	Track-specific
16	ArteraAI Breast	Enterprise (ordered)	Case study	Medium
17	Ibex Prostate Detect	Enterprise	Case study	Track-specific
18	DoseMeRx	Enterprise	Case study	Medium
19	MIM Contour ProtégéAI+	Enterprise	Case study	Track-specific
20	AI-Rad Companion RT	Enterprise	Case study	Track-specific
21	Varian Smart Segmentation	Enterprise	Case study	Track-specific

Suggested Class Structure (if teaching all 21 in one extended session)

- **Hour 1:** The 6 free/open tools — live hands-on lab with #3, #5, #6 especially (highest general relevance)
- **Hour 2:** The research-stage tools (#7-11) as case studies, with #10 (I3LUNG) as the centerpiece HITL discussion
- **Hour 3:** The enterprise tools (#12-21), grouped by track relevance — spend the most time on whichever specialty tools match your actual student cohort (e.g., skip deep radiotherapy detail if no radiation oncology students are present)